GigaWhat Energy

GigaWhat Gas
Controlled procedure

PR-GAS-021 Permit-to-Work for Gas Operations

Field	Value
Document ID	PR-GAS-021
Version	5
Effective from	2025-03-03
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Owner	Gas Safety Manager
Approved by	Head of Gas Network Operations
Applies to	All GigaWhat Gas sites and all gas asset classes

FICTIONAL DOCUMENT. Created for the GigaWhat demo. Not for real operational use.

1 Purpose

This procedure sets out the GigaWhat Gas permit-to-work system. It makes sure that hazardous work on gas plant is planned, that isolations are made and proved before work starts, that everyone knows the limits of the work, and that plant is returned to service safely.

2 Scope

A permit-to-work is required for the following work on any GigaWhat Gas site or asset:

- any work that breaks containment on gas plant, mains or services, whether live or isolated
- maintenance or testing of the pressure reduction station PRS-GWT-01, district governors, slam-shut valves and filters
- purging and commissioning of mains under PR-GAS-031
- hot work in a hazardous area under PR-GAS-040
- entry into a confined space, including governor pits and valve chambers
- mechanical excavation within 3 m of the IP main MAIN-IP-01
- changes to telemetry or control settings that can affect network pressure

A permit is not required for routine inspection that does not break containment (PR-GAS-058), leakage survey (PR-GAS-045) or the first make-safe actions at a reported escape (PR-GAS-003). Any repair that follows a reported escape needs a permit.

3 References

- PR-GAS-003 Response to Reported Gas Escapes
- PR-GAS-031 Purging and Commissioning of Polyethylene Mains
- PR-GAS-040 Working in Hazardous Areas (DSEAR Zones)
- PR-GAS-052 Safe Excavation Near Live Gas Mains
- PR-CORP-001 Reporting Incidents and Near Misses
- PR-CORP-004 Lone Working
- HSE guidance HSG250 (permit-to-work systems) and HSG253 (safe isolation of plant and equipment)
- Gas Safety (Management) Regulations 1996 and HSE guidance L80
- Pressure Systems Safety Regulations 2000 and HSE guidance L122
- Dangerous Substances and Explosive Atmospheres Regulations 2002 (DSEAR) and HSE guidance L138
- Management of Health and Safety at Work Regulations 1999

4 Definitions

Permit-to-work: A numbered written authority, issued by an Authorised Person (Gas), that defines the work, the plant, the isolations, the precautions and the time the permit is valid.

Associated certificate: A certificate issued with a permit for a specific hazard: an isolation certificate, a hot work certificate or a confined space entry certificate.

Permit holder: The Competent Person (Pressure Systems) who accepts the permit and is in charge of the work party.

Positive isolation: Isolation by a spade, blank or physical disconnection, so that gas cannot reach the work even if a valve passes.

Proved valve isolation: Isolation by two closed valves with the section between them vented to atmosphere and watched for pressure rise (double block and bleed).

Suspension: Temporary stopping of work with the isolations left in place.

Cancellation: Formal closure of the permit after the plant is ready to return to service.

5 Responsibilities

Gas Safety Manager: Owns this procedure, appoints Authorised Persons (Gas) in writing, and audits at least 10% of cancelled permits each month.

Head of Gas Network Operations: Makes sure there are enough Authorised Persons (Gas) on duty and on call to issue permits without delay.

Authorised Person (Gas): Plans the isolations, issues and cancels permits and certificates, checks isolations on site personally, and releases each hold point.

Competent Person (Pressure Systems): Accepts the permit, briefs the work party, keeps the work within the permit limits and hands the permit back.

Gas Network Controller: Agrees any valve operation that affects supply, keeps the permit register and records each issue, suspension and cancellation.

6 Procedure

6.1 Planning the work

6.1.1 Define the work, the asset IDs and the boundaries of the plant to be worked on.

6.1.2 Complete a risk assessment and method statement for the task.

6.1.3 Prepare an isolation plan following HSG253, marking every valve, vent and spade position on a sketch.

6.1.4 Check the hazardous area classification under PR-GAS-040 and decide whether a hot work or confined space certificate is needed.

6.1.5 Ask the Gas Network Controller to confirm the effect of the isolation on supply and agree a time for the work.

6.1.6 Check whether other permits are in force on the same plant and coordinate them so that isolations do not conflict.

6.2 Isolation standards

> WARNING: A single closed valve can pass gas without warning. Working behind an unproved isolation can release gas into the work area and cause fire, explosion or asphyxiation.

6.2.1 Use positive isolation for hot work, for confined space entry and for all work on IP plant.

6.2.2 Use at least proved valve isolation for all other work that breaks containment on MP or LP plant.

6.2.3 Use single valve isolation only on LP plant, only with the Authorised Person (Gas) recording why on the permit, and only with continuous gas monitoring at the work.

6.2.4 Lock every isolation valve in position and fit a tag showing the permit number.

6.2.5 Prove each isolation by venting to zero and watching the gauge for 5 minutes on LP plant, or 10 minutes on MP and IP plant, with no pressure rise.

6.3 Issuing the permit

6.3.1 The Authorised Person (Gas) visits the site and checks every isolation, lock and tag personally.

6.3.2 Complete the permit with the permit number, asset IDs, work description, isolations, hazards and precautions, gas test results, associated certificates and validity period.

6.3.3 Set a validity period of no more than 12 hours, ending no later than the end of the permit holder's shift.

6.3.4 Brief the permit holder on site, then the permit holder reads, signs and accepts the permit.

6.3.5 The permit holder briefs every member of the work party, who each sign the permit.

6.3.6 Tell the Gas Network Controller the permit number and time of issue for the permit register.

> HOLD POINT: No work may start until the Authorised Person (Gas) and the permit holder have both signed the permit and the Gas Network Controller has entered it in the permit register.

6.4 During the work

6.4.1 Keep the permit and associated certificates at the work site, in view.

6.4.2 Keep continuous gas monitoring at the work wherever containment is broken.

6.4.3 Stop work and contact the Authorised Person (Gas) if the work needs to go beyond the permit, or if conditions change.

6.4.4 Never alter, remove or operate an isolation under someone else's permit.

6.4.5 Follow PR-CORP-004 if any member of the work party works alone.

6.5 Suspension and shift handover

6.5.1 Suspend the permit at the end of each shift, or when the work is paused for more than 2 hours.

6.5.2 Leave all isolations locked and tagged, and make the work site safe.

6.5.3 Record the suspension on the permit and tell the Gas Network Controller.

6.5.4 Before work restarts, the Authorised Person (Gas) re-checks the isolations, repeats the gas tests and re-issues the permit to the incoming permit holder.

6.6 Hand-back and cancellation

6.6.1 The permit holder confirms on the permit that the work is complete, the work party is withdrawn and tools and materials are removed.

6.6.2 The Authorised Person (Gas) inspects the plant and confirms it is ready to be de-isolated.

> HOLD POINT: Do not remove any lock, spade or isolation until the Authorised Person (Gas) has accepted the hand-back and signed the permit.

6.6.3 The Authorised Person (Gas) supervises de-isolation, purging under PR-GAS-031 where needed, and recommissioning.

6.6.4 Test every disturbed joint at operating pressure before the plant is returned to service.

6.6.5 Cancel the permit and associated certificates, and tell the Gas Network Controller the time of cancellation.

6.7 Emergencies during permitted work

6.7.1 If gas escapes, fire breaks out or anyone is hurt, stop work, make the area safe and withdraw the work party.

6.7.2 Call 999 for fire or injury, and report gas escapes to the Gas Network Controller, or to the National Gas Emergency Service on 0800 111 999 if the Controller cannot be reached.

6.7.3 Treat the permit as suspended until the Authorised Person (Gas) has reassessed the work.

6.7.4 Report the event under PR-CORP-001.

7 Records

Permits, associated certificates and isolation sketches are kept for 6 years after cancellation, and with the asset file where the work changed the plant. The Gas Network Controller's permit register records each permit number, asset, holder, issue, suspension and cancellation times. Monthly audit results are reported to the Head of Gas Network Operations.

8 Revision history

Version	Date	Change
1	2010-03-01	First issue.
2	2013-03-04	Added hot work and confined space certificates.
3	2017-03-06	Isolation standards aligned with HSG253; added proved valve isolation.
4	2021-03-01	Added Gas Network Controller permit register and suspension rules. Mechanical excavation within 3 m of MAIN-IP-01 brought into scope.
5	2025-03-03	Added changes to telemetry and control settings to scope. Limited single valve isolation to LP plant. Added monthly audit by the Gas Safety Manager.